STAT 545 Hw 9

Bowen Zheng

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Problem 1

(CauchyMLE).Consider i.i.d. Cauchy(μ,1) distributed data $X_1, \ldots, X_n$, with density

$$\frac{1}{\pi}[\frac{1}{1+(x-\mu)^2}]$$

A.

We are interested in the maximum likelihood estimate of μ. Derive the Fisher scoring algorithm for finding the MLE.

We simplify a bit from our likelihood equation:

$$f(x_i|\mu) = \frac{1}{\pi}[1+(x_i-\mu)^2]^{-1}$$

The log-likelihood is:

$$\ell(\mu) = \sum_{i=1}^{n} \ln f(x_i|\mu) = -n\ln\pi - \sum_{i=1}^{n} \ln(1+(x_i-\mu)^2)$$

We take the derivative with respect to μ:

$$S(\mu) = \frac{\partial \ell(\mu)}{\partial \mu} = -\sum_{i=1}^{n} \frac{\partial}{\partial \mu} \ln(1+(x_i-\mu)^2)$$

$$S(\mu) = -\sum_{i=1}^{n} \frac{1}{1+(x_i-\mu)^2} \cdot 2(x_i-\mu)(-1) = \sum_{i=1}^{n} \frac{2(x_i-\mu)}{1+(x_i-\mu)^2}$$

Now that we have this, we can easily find the Fisher information:

$$I_1(\mu) = E\left[\left(\frac{\partial}{\partial \mu} \ln f(X|\mu)\right)^2\right]$$

The expected value is the integral inside times the distribution itself:

$$\int_{-\infty}^{\infty} \left(\frac{2(x-\mu)}{1+(x-\mu)^2}\right)^2 \frac{1}{\pi(1+(x-\mu)^2)} dx = \frac{1}{2}$$

Then, we observe n number of X's:

$$I(\mu) = n \cdot I_1(\mu) = \frac{n}{2}$$

The idea from here is that we use Newton's method to finding the MLE using the Fisher Scoring (from notes):

$$\mu^{(k+1)} = \mu^{(k)} + I^{-1}(\mu^{(k)})S(\mu^{(k)})$$

Plug in our values to find:

$$\mu^{(k+1)} = \mu^{(k)} + \frac{\sum_{i=1}^{n} \frac{2(x_i - \mu^{(k)})}{1+(x_i-\mu^{(k)})^2}}{n/2} = \mu^{(k)} + \frac{4}{n}\sum_{i=1}^{n} \frac{x_i - \mu^{(k)}}{1 + (x_i - \mu^{(k)})^2}$$

Our algorithm would be to pick a good starting value of $\mu^{(0)}$ and then updating $\mu^{(k+1)}$ until the norm of the gradient of the change in likelihood is less than some specified tolerance.

B.

> Set true $\mu = 5$. Generate 50 simulated data points from the Cauchy density in part A (you can use rcauchy() in R). Implement your scoring algorithm in R with a starting value $\mu_0 = 0$. Use a tolerance of 10^{-5} to terminate your search. Prepare a table or figure of log-likelihood values against iteration number before the search terminates. Use seed=123 in R so your results can be replicated.

We run the algorithm we derived in the previous step.We can run the simplified final equation but here I just simply plug in our values to the Newton's method:

$$\mu^{(k+1)} = \mu^{(k)} + I^{-1}(\mu^{(k)})S(\mu^{(k)})$$

```r
set.seed(123)

# Generate Data
n <- 50
true_mu <- 5
x <- rcauchy(n, location = true_mu, scale = 1)

# Define Log-Likelihood Function
log_likelihood <- function(mu, data) {
  -length(data) * log(pi) - sum(log(1 + (data - mu)^2))
}

# Define Fisher Scoring Algorithm
mu_curr <- 0
tol <- 1e-5
max_iter <- 1000
history <- data.frame(iteration = 0, mu = mu_curr, log_lik = log_likelihood(mu_curr, x))

# Finally our algorithm
for (i in 1:max_iter) {
  # Calculate Score function S(mu)
  score <- sum(2 * (x - mu_curr) / (1 + (x - mu_curr)^2))
```

```r
  # Fisher Information (I(mu) = n/2)
  fisher_info <- n / 2

  # Update mu
  mu_next <- mu_curr + (score / fisher_info)

  # Record history
  log_lik_next <- log_likelihood(mu_next, x)
  history <- rbind(history, c(i, mu_next, log_lik_next))

  # Check convergence
  if (abs(mu_next - mu_curr) < tol) {
    break
  }

  mu_curr <- mu_next
}

# View the final results
print(history)
```

```
##    iteration        mu   log_lik
## 1          0 0.0000000 -221.8896
## 2          1 0.6451397 -210.8631
## 3          2 1.3634635 -197.2441
## 4          3 2.1681726 -179.7733
## 5          4 3.1085022 -155.9206
## 6          5 4.1556628 -131.0006
## 7          6 4.9409242 -121.3949
## 8          7 5.0549703 -121.2424
## 9          8 5.0482550 -121.2418
## 10         9 5.0485808 -121.2418
## 11        10 5.0485648 -121.2418
## 12        11 5.0485655 -121.2418
```

Wow that sure converges fast.

Problem 2

(Affine invariance of Newton's method). Let $f : R^n \to R$. Consider a Newton iteration to minimize f, given by,

$$x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

Consider a transformation $y = Ax + b$, for some known non-singular $A \in R^n$ and some non-zero $b \in R^{n \times n}$. Show that the Newton step in the y scale is independent of A and b.

Given A non-singular, then A^{-1} exists. Then we define $x = A^{-1}(y - b)$. Then a function of x can be defined as a function of y:

$$f(x) = f(A^{-1}(y - b)) = g(y)$$

Ok, now we can compute the gradient and Hessian of $g(y)$ so that we can use it in Newton's iteration on y. Note: $\frac{dx}{dy} = A^{-1}$. Then,

$$\nabla g(y) = (A^{-1})^T \nabla f(x) = A^{-T} \nabla f(x)$$

$$\nabla^2 g(y) = \frac{d}{dy}[A^{-T} \nabla f(x)] = A^{-T} \nabla^2 f(x) \frac{dx}{dy} = A^{-T} \nabla^2 f(x) A^{-1}$$

Now we use Newton's iteration on y:

$$y_{k+1} = y_k - (\nabla^2 g(y_k))^{-1} \nabla g(y_k)$$

Which is:

$$y_{k+1} = y_k - (A^{-T} \nabla^2 f(x_k) A^{-1})^{-1} (A^{-T} \nabla f(x_k)) = y_k - (A(\nabla^2 f(x_k))^{-1} A^T)(A^{-T} \nabla f(x_k)) = y_k - A(\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

Now, convert everything back to in terms of x (try to get form $\mu^{(k+1)} = \mu^{(k)} + I^{-1}(\mu^{(k)}) S(\mu^{(k)})$). We just replace $y_i = Ax_i + b$:

$$y_{k+1} = Ax_{k+1} + b = Ax_k + b - A(\nabla^2 f(x_k))^{-1} \nabla f(x_k) \implies x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$$

This means that the iteration derived from optimizing y is the same procedure as the iteration derived from doing it on x. Thus, Newton step is independent of the affine transformation.